

Such symmetric exact ground state $|E_0\rangle$ often has the peculiar property that the relative fluctuation of a macroscopic observable $\hat{A}$, which is mostly the order operator $\hat{O}$, does not vanish even in the thermodynamic limit:

$$\lim_{N \rightarrow \infty} \frac{\sqrt{\langle E_0 | \hat{A}^2 | E_0 \rangle - \langle E_0 | \hat{A} | E_0 \rangle^2}}{N} \neq 0,$$

where N is the number of total sites (alias the volume of the system). Since this means that a macroscopic observable does not have a definite value even in the thermodynamic limit, $|E_0\rangle$ is an anomalous state [5] from the view point of thermodynamics where any macroscopic observable is supposed to have definite value [5, 27]. In terms of index p , $|E_0\rangle$ has $p = 2$ and therefore contains superposition of macroscopically distinct states. In other words, the symmetry of the ground state is “obscured” by such large quantum fluctuation. This effect is often called “obscured symmetry breaking” [28].

However, when the exact ground state has such anomalous property, it is often the case that the energy gap between the exact ground state and the low-lying eigenstates decays very fast as $N \rightarrow \infty$ [28, 29]. Then, it is physically allowed to take a linear combination of the exact ground state and some of the low-lying eigenstates to form an approximate ground state $|E'_0\rangle$. Thus constructed $|E'_0\rangle$ are believed to break the symmetry and be “ergodic” in the sense that

$$\lim_{N \rightarrow \infty} \frac{\sqrt{\langle E'_0 | \hat{A}^2 | E'_0 \rangle - \langle E'_0 | \hat{A} | E'_0 \rangle^2}}{N} = 0$$

for any macroscopic observable $\hat{A}$ [28, 29]. In terms of index p , this means that $|E'_0\rangle$ has no superposition of macroscopically distinct states since $p < 2$.

Indeed, Horsch and Linden [29] introduced the trial state $\hat{O}|E_0\rangle$ which approximates the first excited state, and showed that the energy gap between the exact ground state and the trial state decays as fast as or faster than $1/N$. They also showed that a linear combination of the exact ground state and the trial state exhibits the desired Z_2 symmetry breaking.

Koma and Tasaki [28] rigorously showed that the linear combination $|\psi\rangle$ of the exact ground state and the trial state is also the ergodic state in the sense that

$$\frac{1}{N^2} \left[\langle \psi | \hat{A}^2 | \psi \rangle - \langle \psi | \hat{A} | \psi \rangle^2 \right] = 0 \quad (9)$$

for any translationally invariant $\hat{A}$.

As is pointed out in Ref. [10], the low-temperature coherence properties of quantum memory is closely related to the obscured symmetry breaking in statistical physics. Indeed, our results in this paper are considered as improvement of the previous results, since Hamiltonian Eq. (2) is equivalent to that of the transverse Ising model [35] in condensed matter physics. (For example, this model was used to describe the order-disorder transition in some double-well ferroelectric systems, such

as potassium dihydrogen phosphate (KH_2PO_4) crystals [36].)

First, by numerical calculations, we have explicitly shown for the first time how the macroscopic coherence properties of the exact ground state changes when the transverse magnetic field is changed (Fig. 1). We have also visualized the structure of the macroscopic superposition in the exact ground state (Fig. 2), and seen that the exact ground state is approximately an equal weight superposition of two symmetry-broken phases.

Second, we have shown that only $\hat{M}_z$ fluctuates macroscopically in the $\lambda < 1$ phase (Fig. 3). Since the ground state is symmetric, this means that the second moment of $\hat{M}_z$ is of $O(N^2)$ in that phase. In terms of statistical physics, this means that $\hat{M}_z$ is the unique order operator in this phase, which is, to the author’s knowledge, a new result.

Third, we have shown that the equal-weight superposition of the exact ground state and the first excited state has $p = 1$ (Fig. 5). This means that the superposition of the exact ground state and the first excited state is ergodic. Although similar results have been obtained, the advantages of our result are (i) instead of the trial state, we have directly used the first excited state to show the ergodicity, (ii) our result that the fluctuation is of $O(N)$ is stronger than Eq. (9), and (iii) we have shown the ergodicity for any additive operator, whereas only translationally invariant additive operators are considered in the previous studies. (A disadvantage of our result is that it is less general since we have used numerical calculations.) In summary, to consider index p for the ground states of many-body Hamiltonians in condensed matter physics is very useful for the study of the foundation of statistical physics.

ACKNOWLEDGMENTS

The author thanks A. Shimizu and Y. Matsuzaki for useful discussions. This work was partially supported by Japan Society for the Promotion of Science.

Appendix

1. Proof of Eq. (7)

Note that $|\Psi\rangle$ is a simultaneous eigenvector of $\hat{M}_x$, $\hat{M}_y$, and $\hat{M}_z$ corresponding to the eigenvalue 0, since a singlet is a simultaneous eigenvector of x , y , and z component of the total magnetization corresponding to the eigenvalue